Denis Maxheimer

AI / ML Engineer • DevOps
Mannheim, Germany · Fully remote worldwide · On-site/Hybrid in Prague
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EU citizen (German) - no work permit required in EU/EEA

PROFILE

Data Science MSc candidate with three years of applied engineering experience across the full data lifecycle — from API ingestion pipelines and PostgreSQL schema design to model training and production web applications.

Current focus on LLM-based systems: agentic workflows and benchmark evaluation methodology. Operating a self-managed Linux server environment with containerised services, reverse proxying, monitoring, and secure networking.

Seeking an AI/ML Engineer or DevOps role where model development and production infrastructure meet. Ready to start January 2027.

EXPERIENCE

Data Engineer & Full-Stack Web Developer *May 2021 - Feb 2023*
Dept. of Water Resources and Environmental Modeling, CULS *Prague, Czechia*
Began as a course project; invited to continue as a paid engineer on the department's environmental monitoring platform.

- Designed and maintained a PostgreSQL database consolidating environmental sensor and public API data from multiple sources
- Built Python ingestion services that automatically retrieved, validated, and persisted data, with duplicate detection and quality checks
- Developed a full-stack web application (Django backend, Vue.js frontend) for data access and visualisation
- Deployed and operated services on Kubernetes with Django, PostgreSQL, Celery, and Redis
- Built a maintenance reporting system used by the research team to track sensor fleet health

Machine Learning Engineer (Python) *Oct 2021 - Jan 2022*
Dept. of Water Resources and Environmental Modeling, CULS *Prague, Czechia*

- Researched and implemented HAN (Holistic Attention Network) for single-image super-resolution
- Trained the model on the project's planetary imagery dataset and evaluated resolution improvement
- Owned the training and evaluation pipeline using PyTorch, OpenCV, and NumPy

Project Researcher *Nov 2020 - Dec 2020*
Dept. of Water Resources and Environmental Modeling, CULS *Prague, Czechia*

- Performed curation and preprocessing of planetary imagery datasets for downstream machine learning use
- Rehired by the department the following year based on performance

Delivery Associate (part-time) *2023 - present*
ABV Saarbrucken *Germany*
Part-time role held continuously while relocating to Germany, completing German language certification, and pursuing an MSc full-time.

EDUCATION

MSc Data Science *2025 - expected 2027*
University of Mannheim *Germany*

- Coursework: Large Language Models & Agents, Generative Software Engineering, Data Mining, Web Mining, Machine Learning, Large Scale Database Management, Advanced Methods in Text Analytics
- Six-month team project: *Evaluation on Live Generated Benchmarks* - methodology for reducing benchmark contamination in LLM evaluation

German Intensive Language Course *2024 - 2025*
Abendakademie Mannheim *Germany*

- Completed with certified B1 proficiency

BSc Environmental Data Science *Sep 2019 - May 2022*
Czech University of Life Sciences *Prague, Czechia*

- Specialisation in Informatics: data structures, storage, and large-scale data handling
- Thesis: *Development of a Web Application for Air Quality Monitoring in the Czech Republic* - grade 2 (very good)

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, R, Bash
ML / AI	PyTorch, scikit-learn, LLM APIs, agentic workflows, OpenCV, NumPy, pandas
Data	PostgreSQL, ETL pipeline design, API ingestion, Celery, Redis, data quality validation
Infra	Docker, Docker Compose, Kubernetes, OpenTofu, Traefik, Ansible, Proxmox, Linux (Ubuntu, Arch), OpnSense, WireGuard, Cloudflare, Git, CI/CD
Web	Django, Vue.js, REST APIs

LANGUAGES

Russian (**native**) · English (**C1**) · Czech (**B1**) · German (**B1**)